93

Ctrl Fn Cmd Alt Lang Alt Home

1

$T = 625\,K \qquad t = 500\text{ days} \qquad t^0 = 600\ \mu m$

17/18

a)

$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{T}\right)$

$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{625}\right)$

$\delta^* = 2.116$

$t^* = 6.62\times10^{-7} \exp\left(\frac{11949}{T}\right)$

$t^* = 6.62\times10^{-7} \exp\left(\frac{11949}{625}\right)$

$t^* = 133$

$t > t^*$

$\delta = \delta^* + K_L(t - t^*)$

$K_L = 7.48\times10^6 \exp\left(\frac{-12500}{625}\right)$

$K_L = .0154$

$\delta = 2.116 + .0154(500 - 133)$

$\boxed{\delta_{(\mu m)} = 7.7178}$

b) $f = 14\%$ $\quad \rho_{Zr} = 6.5\ g/cc$

$PBR = 1.56$ $\quad \rho_{ZrO_2} = 5.68\ g/cc$

$$C_H^{clad}(wtppm) = \frac{2f \times \delta \times \rho_{oxide} \times f^0_{ZrO_2} \times \frac{M_H}{M_O} \times 10^6}{\left(t - \frac{\delta}{PBR}\right) \times \rho_{metal}}$$

$f^0_{ZrO_2} = \frac{\cancel{16}\ 32}{\cancel{123}.22491} = .13 \Rightarrow 0.26$

$\frac{M_H}{M_O} = \frac{1}{16}$

$\cancel{C_H^{clad}(wppm) = \frac{2(.14) \times 7.7178 \times 5.68\,g/cc \times .13 \times \frac{1}{16} \times 10^6}{(500 - 7.7178)}}$

$$C_H^{clad}(wppm) = \frac{2(.14) \times 7.7178\,\mu m \times 5.68\,g/cc \times .13 \times \frac{1}{16} \times 10^6}{\left(600\,\mu m - \frac{7.7178\,\mu m}{1.56}\right) \times 6.5\,g/cc}$$

$$\frac{100375.5116}{5867.634} = 25.95\ wppm$$

$\times 2$

2/2

(2) The rate limiting step in the aqueous corrosion of Zr cladding is the diffusion of oxygen through the oxide.

8/8

(3) Hydrides form in the rim of the cladding.

Hydrides preferentially form in lower temperature regions with tensile stress

Hydrides reduce fracture toughness, increase brittleness, increase oxidation rate, increase irradiation growth

11/12

(4) Reactivity insertion accident.

PWR

Control rod becomes mechanically detached from the drive mechanism. Coolant pressure ejects rod out of the core, Reducing absorber material in that part of the core leading to power and temperature spikes.

BWR

Control blade becomes detached from drive mechanism. blade gets stuck in place. Blade eventually drops out of core causing reactivity, power, and temperature spike.

Rod is ejected from core → Reactivity goes up → Power goes up → Temperatures go up.

The enthalpy of the fuel rises quickly.

The changes in power and temperature can cause excess stress on the cladding causing rupture, and release of fission products and fuel into coolant. Increased temperatures could also melt fuel.

The cladding could bow during this event blocking passage of coolant leading to a DNB scenario

-wanted a little more

12/12

(5) Loss of coolant accident. A rupture in a pipe or pressure vessel causes a sudden loss in coolant and pressure. Activates ECCS and initiates a scram. The fuel is still producing heat and the drop in coolant pressure can lead to a pressure differential between the rod and the coolant causing cladding ballooning and burst. The injection of coolant by the ECCS can also cause a thermal shock that fractures the cladding. In a LOCA the rupture of the cladding is due to pressure differences and/or the coolant shock where in a RIA the cladding fails due to the power spike in the fuel leading to PCI induced failures.

4/4

(6) With higher burnups the cladding is more brittle due to hydrogen pickup, oxidation, and irradiation hardening. Failures are more likely to occur at higher burnups, and the failure mechanism is more likely to be fracture.

4/6

(7)
- increase melt margin
- reduce hydrogen production
- reduce oxidation rate
- retain more fission products in fuel

> not really... related, but more general...

6/6

(8) Doped fuel. Doped fuel is of interest because it changes the oxygen diffusion in the fuel, leading to larger grain sizes during sintering and an increase in fission product retention.

5/6

(9) When zirconium cladding is exposed to high temperature steam the rate of oxidation increases. Since oxidation is exothermic, it continues increasing the temperature of the cladding, increasing oxidation. This process also releases hydrogen into the pressure vessel. This runaway oxidation can eventually cause the cladding to fracture or excess hydrogen buildup

- phase changes..

(10) – Oxidation. Total oxidation must be limited to < 17% of clad thickness.

– Liftoff – Separation of clad and fuel will increase fuel temperature, increasing FGR and rod internal pressure. Requirement is clad creepout cannot exceed fuel swelling/expansion

– Cladding wear – cladding wear cannot reduce cladding thickness by more than 10%.

6/6

(11) Crud will get deposited on the surface of the fuel rod. Crud has a low thermal conductivity, increasing fuel temperatures. In PWR's crud can also contain boron, locally reducing reactor power. Crud also contains radioactive materials such as Co-60 which is of concern during outages to plant personnel

6/6

(12) – In PWR's LiOH is used to maintain pH > 6.4. This reduces corrosion of the cladding, and as boron concentration is reduced LiOH concentration is reduced to maintain pH.

6/6

– noble metal injection is used to prevent the adherence of crud. This prevents radioactive species like Co-60 from adhering to plant components, keeping them in the coolant longer where they will eventually be removed by the cleanup systems.

(13) – once in operating conditions, the metal in the MOX will begin to evaporate in the fuel. This causes a movement down the thermal gradient towards the cooler side of the fuel. The voids move towards the hotter center of the fuel, where a central void forms.

– pore migration @ high T

6/8

– The MOX fuel grains reorient themselves into several distinct zones where there is a central void, then columnar grains, then equiaxed grains, then as-sintered grains at the rim.

– two related aspects of one phenomenon